

Logic Bubble

A Visual Logic-Mapping Web Application

IMM Web App Delivery
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Project Overview

What is Logic Bubble?

- A web application for visual logic mapping
- Converts text into interactive diagrams
- Users extract key ideas as “bubble nodes”
- Connect ideas using directional arrows

Designed for understanding complex information more easily



Problem

第 1 项

Difficult to track relationships in long or complex text

第 2 项

Mental models are hard to maintain

第 3 项

Existing tools are:

- Too complex (Miro, Lucidchart)
- Not designed for text extraction

No simple tool exists for turning text into visual logic structures

Solution

Our Solution

- Paste text into the editor
- Extract key phrases as bubble nodes
- Visually connect ideas on a canvas
- Save and revisit projects anytime

Key Value:

- Lightweight
- Browser-based
- Focused workflow



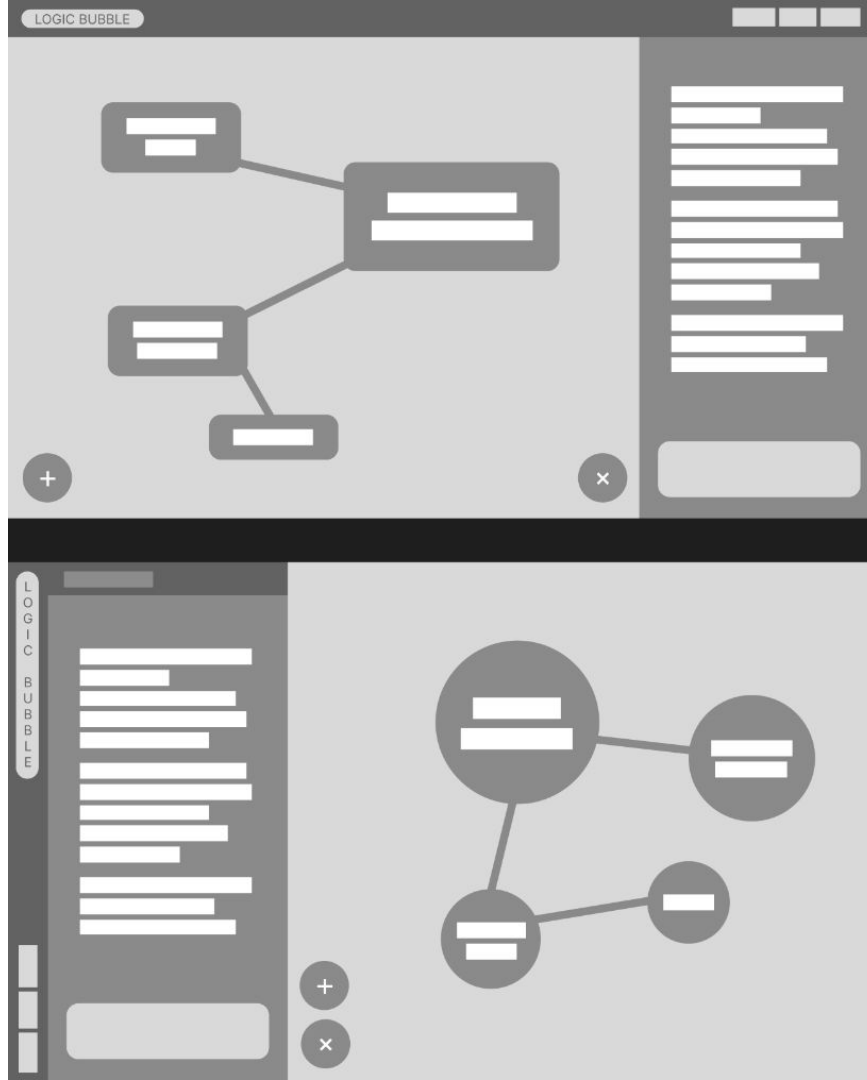
Target Users

Who is it for?

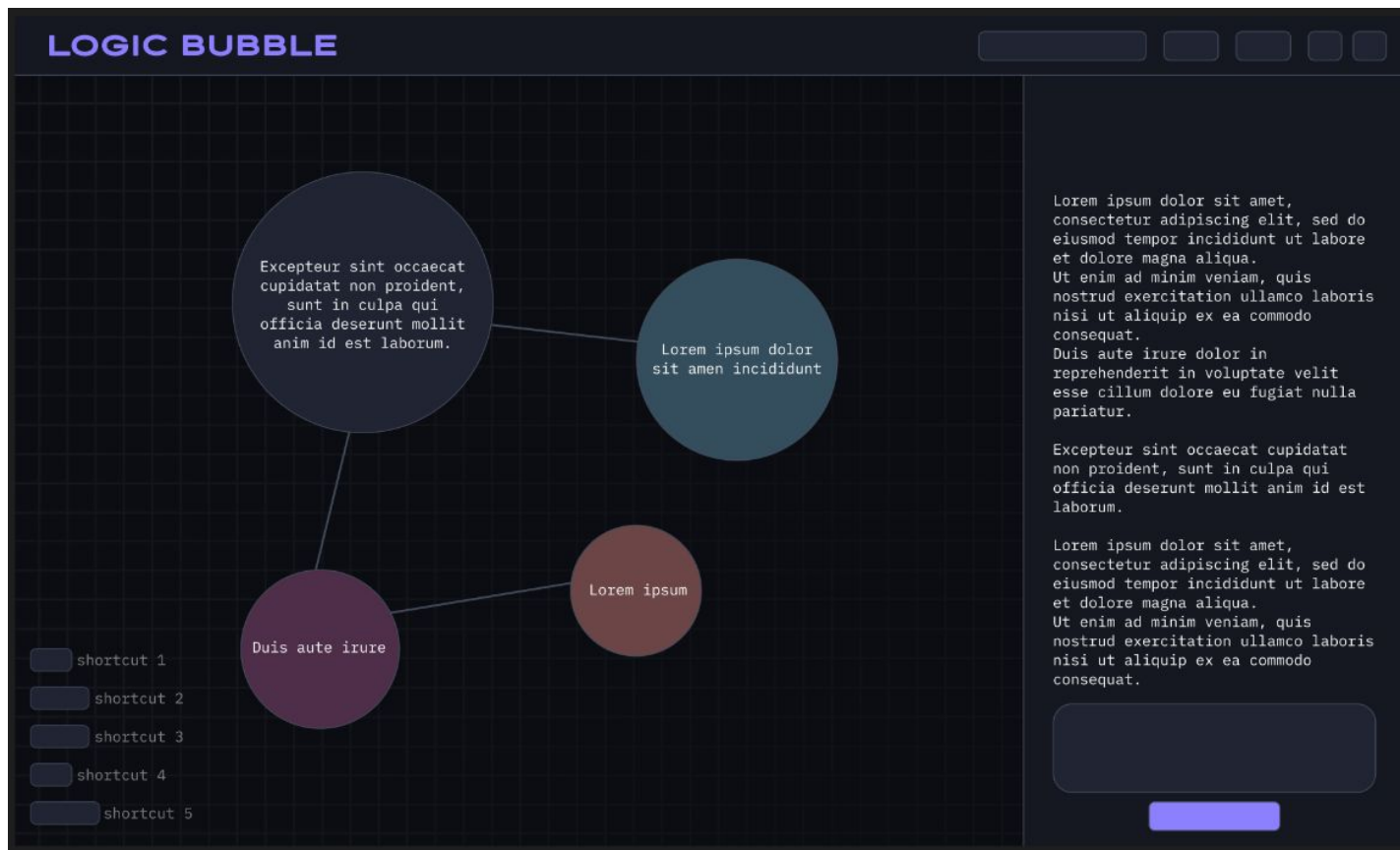
- Students (study diagrams, theory mapping)
- Researchers (analyzing documents)
- Writers (tracking narrative structure)
- Knowledge workers (organizing ideas)

Designed for users working with
complex, text-heavy content

Sketches



Wireframe





Key Features

Core Features

- Create and edit bubble nodes
- Drag and position freely on canvas
- Connect nodes with arrows
- Insert dot nodes for structure
- Undo system (up to 80 steps)
- Export as PNG or JSON

Full interactive editor with keyboard shortcuts



User System & Projects

User & Project Management

- User registration and login
- Session-based authentication
- Save projects with titles
- Dashboard to manage projects
- Edit, export, and delete projects

All data stored in MySQL database



System Architecture

Frontend

- HTML + CSS (Tailwind)
- Vanilla JavaScript (ES Modules)

Backend

- Python Flask

Database

- MySQL

No build tools — simple and efficient architecture



Application Structure

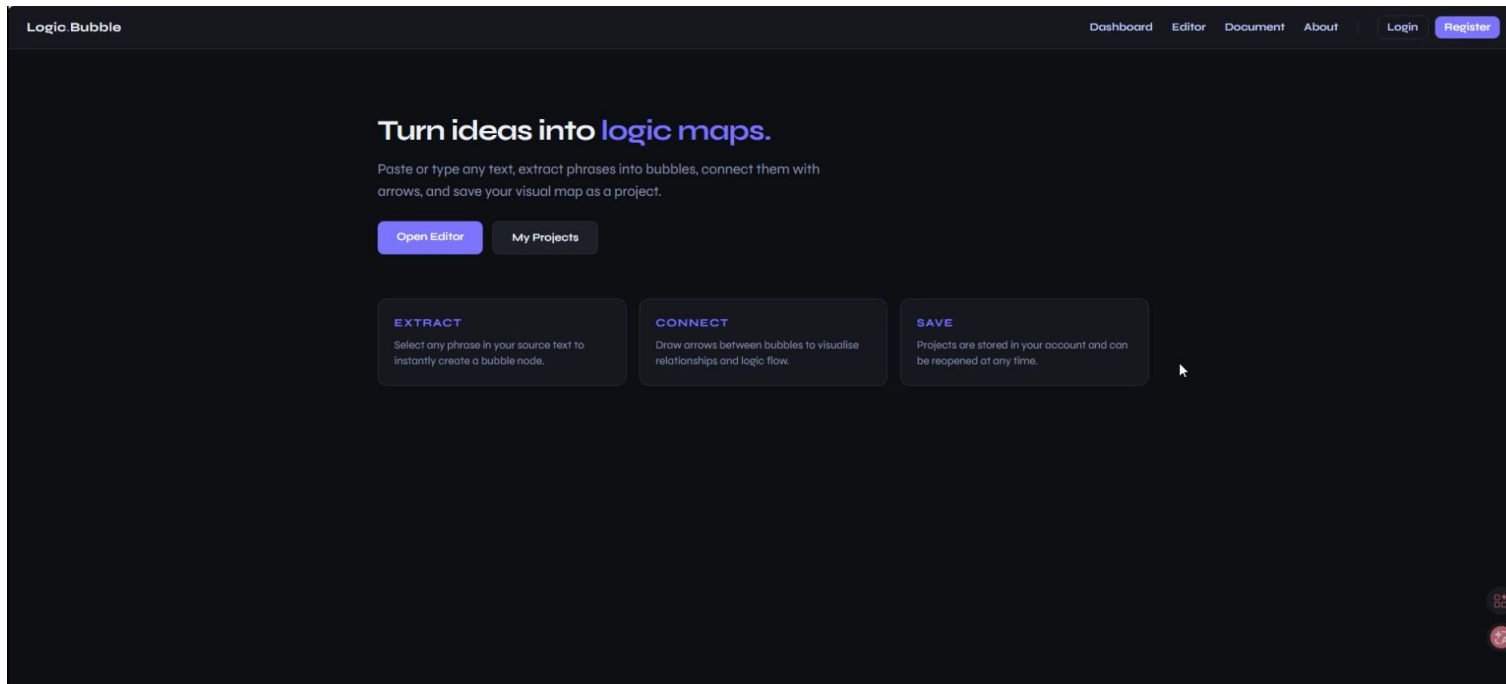
Main Pages

- Home (entry point)
- Login / Register
- Dashboard (project management)
- Editor (main workspace)
- Documentation & About

Clear separation between public and authenticated pages



Final Product



LOGIC BUBBLE

CONNECT

Untitled Project

Save

File

Edit

?

o

Web Design

interactivit

topic

WordPress

FILL COLOR

Opacity

25%

Apply

Clear

SOURCE TEXT

SELECT TEXT + ENTER TO EXTRACT

Sony kindergarten
Instructions
Group size: 3-4

If client projects are not feasible,

Using your research skills, knowledge of Web Design and web technologies and your creativity, create an application that could be used in the real world. Think about a product that you can develop for other people to use.

You are free to choose your own topic. It must go beyond an informational website and must provide user-friendly interactivity.

All of the following technologies must be used: HTML5, JavaScript, CSS, PYTHON/MYSQL (or node + mongo). Other technologies should be approved by the lecturer.

Please make use of libraries, frameworks, design systems, best practices (including accessibility and security) and all tools available to you (webpack, CI or Automated tests for example). You must adhere to the licenses and authors must be properly attributed.

WordPress is NOT allowed for this project.

The application should be deployed on a public server. If this is not possible, please provide instructions on how to run the application on a local machine.

If your application will be deployed for mobile devices, please arrange to install it on the test device that is accessible to the lecturer.

Grading:

INPUT TEXT

Paste or type your source text here...

Watch Video Demo



Development Process

Workflow

- Agile-inspired development
- 4 structured sprints:
 - Research & Design
 - Core Build
 - Feature Development
 - Bug Fix & Polish

Iterative development with continuous testing



Challenges

Challenges Faced

- Designing canvas data structure (graph JSON)
- Managing complex UI interactions
- Synchronizing frontend and backend
- Fixing bugs during feature expansion



Future Improvements

Future Plans

- AI-powered diagram generation
- PDF import and annotation
- Light theme support
- Custom font upload
- Embedded use in reading tools

Expanding functionality while keeping simplicity



Conclusion

Summary

- Logic Bubble simplifies complex information
- Combines text and visual thinking
- Lightweight and easy to use
- Strong foundation for future expansion

Thank You!

Questions?

